

A 150-GBaud PAM2, 120-GBaud PAM4 Monolithic Transmitter Integrating a Silicon-Photonic Mach–Zehnder Modulator and a CMOS Driver

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Abstract We demonstrate a monolithically integrated optical transmitter consisting of a 46-GHz silicon photonic Mach-Zehnder modulator and a CMOS-SOI driver with on-chip continuous-time linear equalization. The measured E/O bandwidth is 70 GHz, with BERs below HD-FEC at 240-Gb/s PAM4 and 150-Gb/s PAM2. ©2026 The Author(s)

Introduction

Evolving Ethernet 800G and 1.6T standardization targets optical interfaces with ≥ 200 Gb/s/lane 4-level Pulse Amplitude Modulation (PAM4) signaling [1]. This trend has driven growing interest in Linear Pluggable Optics (LPO) for short-reach optical interconnects, owing to its low-power architecture and reduced latency. However, by reducing reliance on module-side digital signal processing (DSP), LPO places more stringent requirements on the optical transmitter, particularly in terms of bandwidth and linearity. To preserve signal integrity at 200 Gb/s/lane PAM4, the transmitter generally requires an effective bandwidth well beyond the 53 GHz Nyquist frequency, with a practical target approaching 70 GHz [2].

To meet this bandwidth requirement, several material platforms have been investigated in recent years. Emerging devices such as Thin-film Lithium Niobate (TFLN) and Silicon-Organic Hybrid (SOH) promise Mach-Zehnder modulators (MZMs) with wide bandwidth and low V_{pi}L, but their cost and readiness for high-volume manufacturing (HVM) for intensity-modulation direct-detection (IMDD) applications remain a topic of active research and development [3-8].

In contrast, silicon photonic (SiP) modulators are attractive for their complementary metal-oxide semiconductor (CMOS) compatibility and strong potential for high-volume manufacturing. SiP Mach-Zehnder modulators (MZMs) are widely used in optical links because of their broadband optical response and ease of bias control. Long MZMs are implemented with traveling-wave (TW) electrodes for broadband electro-optic (E/O) response [9-11]. SiP TW-MZMs have demonstrated a -3dB E/O bandwidth up to 70 GHz in silicon-on-insulator (SOI) photonic processes, requiring separate driver (DRV) ICs in a multi-chip assembly [12]. Co-design approaches using CMOS or SiGe drivers and driver-side passive networks, such as T-coils, to compensate for the high-frequency electrode

loss of TW-MZMs and extend the E/O bandwidth have also been reported [13-15]. However, for CMOS DRV's driving SOI photonics, the reported overall bandwidth has not exceeded 65 GHz, while the in-band response flatness of existing implementation remains limited, thereby constraining high-speed signal quality [14]. A monolithic SiP process reduces the overall cost in comparison to an assembly of a separate electronic IC (EIC) and photonic IC (PIC), when the cost associated with packaging, testing and validation is taken into account.

In this paper, we present a broadband monolithic SiP transmitter (TX) in the GlobalFoundries 45SPCLO process, achieving an overall bandwidth exceeding 70 GHz in a low-cost, low-area implementation for LPO applications, as shown in Fig. 1. It consists of a driver (DRV) in CMOS-SOI integrated with a TW-MZM available in the process design kit (PDK). A micrograph of the transmitter (TX) is depicted in Fig. 2(a). This performance is enabled by a new flipped-polarity transformer architecture combined with T-coil peaking for driver-side continuous-time linear equalization (CTLE). Through tight E/O co-design, the driver frequency response is peaked at 67 GHz to effectively

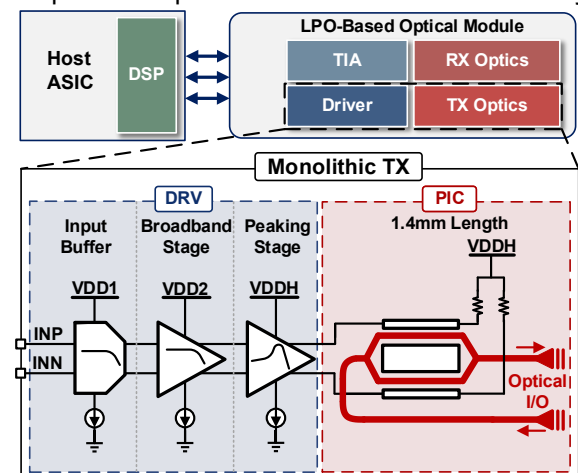


Fig. 1: Simplified LPO system and SiP TX architecture

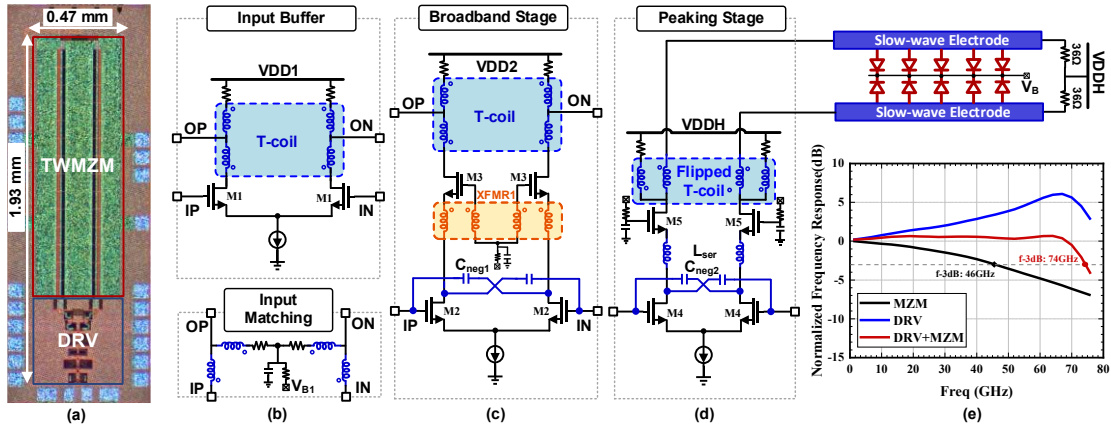


Fig. 2: (a) Micrograph of the monolithic prototype. (b) Schematic of the input stage, (c) broadband stage, and (d) peaking stage. (e) Simulated frequency response of MZM, DR, and overall TX front-end.

compensate the 46-GHz 1.4 mm TW-MZM. The TX front-end demonstrates 240 Gb/s PAM4 and 150 Gb/s PAM2 operation with BER below the HD-FEC threshold. The measured energy efficiency is 1.68 pJ/bit and 2.68 pJ/bit, respectively, with an RLM of 0.92 in PAM4 mode. To the best of our knowledge, this is the first demonstration of a 240 Gb/s PAM4 SiP MZM-based TX in a monolithic CMOS-SOI driver.

MZM Characterization and E/O Co-design

The SiP TW-MZM is implemented on a 300-mm SOI wafer with a 160-nm top-silicon layer and a 2- μ m buried oxide [16]. Fig. 2(a) shows the micrograph. A coplanar-waveguide (CPW) transmission line with a GSGSG configuration supports differential driving, thereby effectively improving the signal swing. To provide a local return path for the differential signal, the three ground metal planes are periodically connected to form a continuous ground plane, which helps preserve the bandwidth. In addition, a T-shaped slow-wave structure is employed to achieve velocity matching. The characteristic impedance of the utilized MZM is 36 Ω . As the reverse bias increases from 1 V to 4 V, the TW-MZM provides a bandwidth from 39.3 GHz to 47.4 GHz, while the $V_{pi} \cdot L$ varies from 1.65 V $\cdot$ cm to 2 V $\cdot$ cm. Considering the trade-off between efficiency and bandwidth, a 3-V reverse bias is adopted in this design to achieve a $V_{pi} \cdot L$ of 1.9 V $\cdot$ cm and a frequency response as shown in Fig. 2(e).

The DRV, consisting of an input buffer, a broadband stage, and an output peaking stage, is designed to provide sufficient output swing and equalization to compensate for the MZM electrode roll-off, as the schematics shown in Fig. 2(b)–(d). Three techniques enable the overall bandwidth beyond 70 GHz. First, negative capacitors (C_{neg1} and C_{neg2}) are introduced in the broadband and peaking stages to partially cancel their input capacitance, thereby relaxing the driving requirement on the preceding stage. The input buffer omits the negative capacitance to

preserve broadband input matching. Second, a new technique leveraging a flipped-polarity transformer (XFMR1) is employed in the broadband stage to enhance the current gain of M3 when driving the heavy capacitive load of the peaking stage. Together with the output T-coil network, this extends the stage bandwidth, maintaining a flat gain response beyond 67 GHz and preventing premature attenuation before the subsequent peaking stage. Third, a flipped T-coil is adopted in the output stage to provide compact shunt and series peaking, generating 3 dB peaking at 67 GHz for CTLE. Combined with the additional 3 dB peaking from the preceding stages, the overall driver achieves 6 dB peaking at 67 GHz. Fig. 2(e) shows the simulated frequency responses of the standalone DRV and the proposed DRV–MZM combination. With monolithic integration, the simulated E/O -3 dB bandwidth increases to 74 GHz.

Testing Setup and Measurement Results

The experimental setup is shown in Fig. 3(a). A PRBS15 pattern is generated by a 256 GSa/s arbitrary waveform generator (AWG) with a 1 Vpp differential output swing. Cable and DC-block losses are compensated by importing their S-parameters into Keysight IQTools. No transmitter-side pre-emphasis is applied in the TX DSP. RC pulse shaping is enabled only for symbol rates above 120 GBaud to account for the AWG's finite sample rate and 80 GHz analog bandwidth, improving waveform fidelity over the transition mode at these rates. The optical carrier is provided by a 1310 nm continuous-wave (CW) laser, amplified by a semiconductor optical amplifier (SOA), and passed through an isolator and a polarization controller (PC). The optical power at the fiber-array (FA) input is 17 dBm. After the chip, the optical signal passes through a Bi-doped fiber amplifier (BDFA), an optical band-pass filter (OBPF), and a Praseodymium-doped fiber amplifier (PDFA). These amplifiers

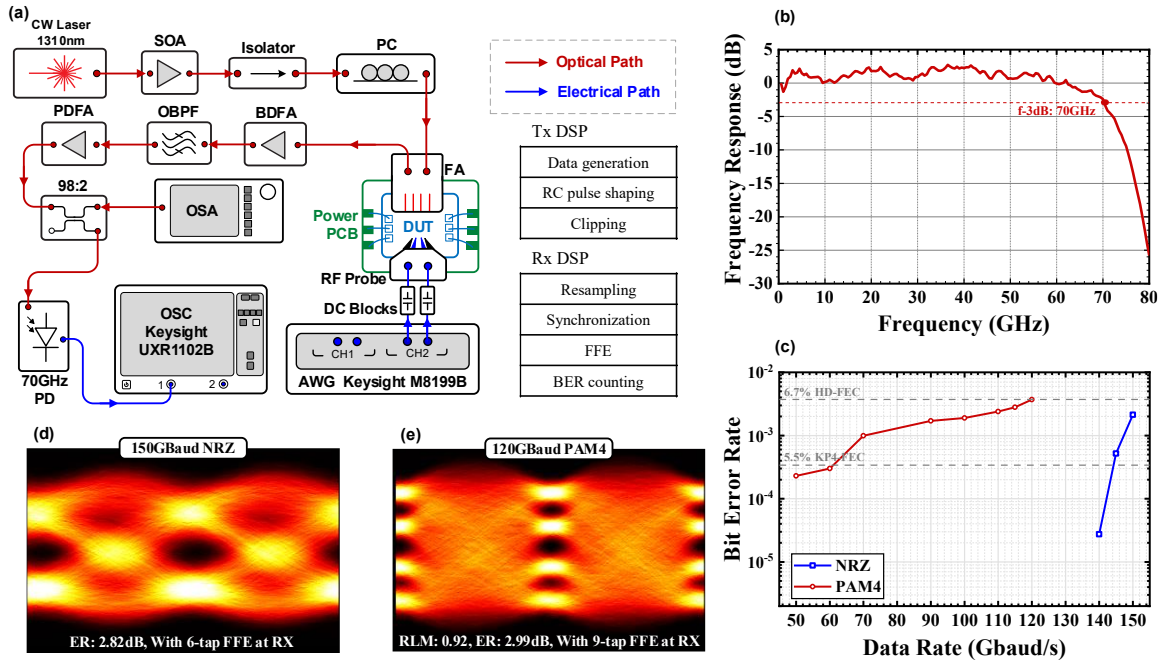


Fig. 3: (a) Testing setup for eye diagram and BER testing. (b) Measured 3-dB E/O bandwidth. (c) Measured BERs of NRZ and PAM4 versus data rate. (d) Eye diagram of 150GBaud NRZ. (e) Eye diagram of 120GBaud PAM4.

compensate for grating-coupler loss and increase the received power, since the photodiode (PD) is used without a transimpedance amplifier. A 98:2 tap routes 2% of the optical power to an optical spectrum analyzer (OSA) for spectrum monitoring. The remaining optical signal, with 4 dBm power, is detected by a 70 GHz photodiode, and the electrical waveform is captured by a 256 GSa/s real-time oscilloscope. The captured waveform is processed offline in the RX DSP for resampling, synchronization, feed-forward equalization (FFE), and bit-error-rate (BER) counting.

Fig. 3(b) shows the measured E/O response of the proposed TX with a -3 dB bandwidth of 70 GHz. Fig. 3(c) shows the measured BER versus data rate for PAM4 and NRZ with 9-tap and 6-tap FFE enabled, respectively. Limited by the oscilloscope record length, the minimum measurable BER is 5.0E-6. For PAM4, a BER of

3.7E-3 is measured at 120 Gbaud, which remains below the 6.7% HD-FEC threshold. For NRZ, the BER remains below the 5.5% KP4-FEC threshold up to 140 Gbaud and at 150 Gbaud, the measured BER is 2.13×10^{-3} , also below the 6.7% HD-FEC limit. Fig. 3(d) presents the measured NRZ eye diagrams at 150 Gbaud with an extinction ratio (ER) of 2.82dB. Fig. 3(e) shows a 120Gbaud PAM4 eye with an ER of 2.99dB and an RLM of 0.92. The total power consumption of the transmitter in this test is 403mW. Fig. 4 summarizes the performance of the state-of-the-art EO-integrated optical transmitters. Among front ends using CMOS-based driver technology, this work achieves the highest data rate.

Conclusions

To the best of our knowledge, we demonstrate the first 240 Gb/s PAM4 SiP MZM TX with a monolithic CMOS-SOI driver and a 70 GHz E/O - 3 dB bandwidth, without any FFE from the AWG. A single monolithic chip saves cost compared to separate EIC-PIC chips. The overall bandwidth can be increased in the future with a custom MZM design.

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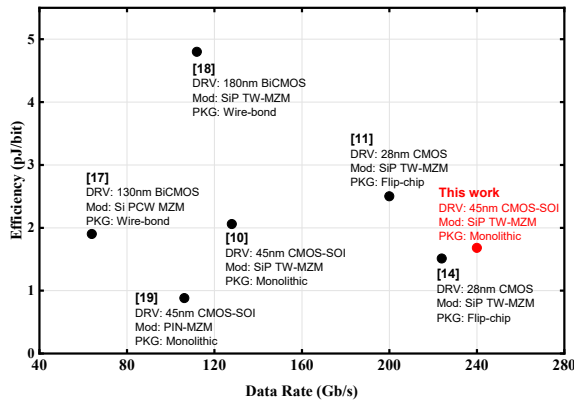


Fig. 4: Comparing state-of-the-art EO-integrated SiP transmitters.

PKG: package, PCW: Photonic crystal waveguides, CMOS: complementary metal-oxide semiconductor.

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